

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

RAPC203: Mechanics of Materials
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define stress	M 1.02	U
2	Define volumetric stress	M 1.04	A
3	What are the different types of beams	M2.01	U
4	Write the relationship between bending moment, bending stress of a beam	M2.03	U
5	Write the torsion equation	M3.01	U
6	Write Euler's equations of buckling	M3.05	A
7	What s spring index?	M3.03	U
8	What is resilience?	M4.01	U
9	Write the name of a composite material	M4.02	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

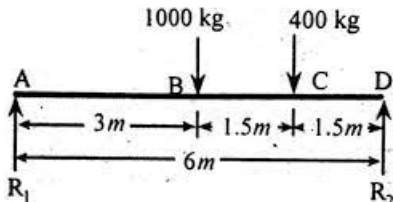
		Module Outcome	Cognitive level
1	Explain any 3 mechanical properties of materials	M 1.04	A
2	Explain hoop stress and longitudinal stress	M 1.05	A
3	Draw the bending moment diagram of a cantilever beam caring point load at the end.	M2.02	A
4	Explain the assumptions of simple bending	M2.03	U
5	What is polar moment of inertia? Explain	M3.01	U
6	Explain equivalent length of a column.	M.3.05	A
7	Explain proof resilience	M4.01	U

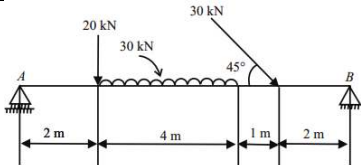
8	Write any three properties of steel	M4.02	U
9	Explain Hooks law	M1.02	U
10	Compare solid and hollow shaft based on strength	M3.01	U

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	A steel rod of 20m length at a temperature of 25°C is heated to 75°C. Modulus of elasticity of steel is 2×10^5 N/mm ² and temperature coefficient of expansion is 12×10^{-6} per °C. Find (i) Thermal stress induced when expansion is completely prevented. (ii) Total expansion prevented.	M1.05	A
OR			
2	A material has modulus of rigidity equal to 0.4×10^5 N/mm ² and bulk modulus equal to 0.75 N/mm ² . Find its Young's modulus and Poisson's ratio.	M1.04	A
3	A simply supported beam having a span of 7m on which a uniformly distributed load of 3kN/m is acting over 4m from the left support. Draw shear force and bending moment diagram.	M2.03	A
OR			
4	Determine the support reactions. 	M2.01	U
5	Calculate the maximum intensity of shear stress induced and the angle of twist produced in degrees in solid shaft of 100 mm diameter, 10 m long transmitting 112.5 kW at 150 rpm. Take $G = 82$ kN/mm ² .	M 3.02	A
OR			
6	Calculate the critical load of a strut which is made of a bar, circular in section and 5m long and which is pin jointed at both ends. The same bar when used as simply supported beam gives a mid span deflection of 10 mm with a load of 10N at the center.	M3.05	A
7	Determine the support reactions.	M 3.01	U

			
OR			
8	Explain Open coiled and close coiled spring on account of design features.	M 3.04	U
9	Draw stress strain diagram of ductile material and identify significant points	M1.02	U
OR			
10	Explain different types of loading applied in beams with neat diagrams.	M 2.01	U
11	Explain the concept of strain energy under different loading conditions	M4.01	U
OR			
12	Explain case hardening of titanium and alloys	M4.02	U

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

RAPC203: Mechanics of Materials
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define hardness of a materials	M 1.01	U
2	What is factor of safety	M1.04	A
3	What are the different types of loading acting on a beam	M2.01	U
4	Write any two assumptions of simple bending	M2.03	U
5	Write the Rankine equation for buckling	M 3.05	A
6	What is stiffness?	M3.04	U
7	What is strain energy?	M4.01	U
8	Write any one mechanical property of stainless steel	M4.02	U
9	What is elasticity?	M1.01	U

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Draw stress -strain curve for ductile and brittle materials	M 1.01	U
2	Explain the relationship between elastic constants	M1.03	A
3	Draw the shear force diagram of a simply supported beam carrying a centre load	M2.02	A
4	Derive the relationship between bending moment, bending stress and radius of gyration of a beam	M2.03	U
5	Explain different types of springs.	M3.02	U
6	Explain torsion equation.	M3.01	U

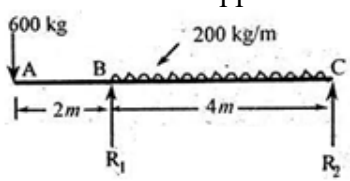
7	Explain modulus of resilience	M 4.01	U
8	Write any three properties of titanium alloys	M4.02	U
9	Explain different types of loadings	M2.01	U
10	Compare open coiled and closed coiled spring?	M3.01	U

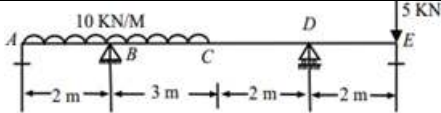
PART C

VI. Answer all questions from the following

(6x 7 = 42 Marks)

Module Outcome Cognitive level

1	A flat steel bar 200mm X 20mm X 8mm is placed between two aluminium bars 200mm X 20mm X 6mm. So as to form a composite bar. All the three bars are fastened together at room temperature. Find the stresses in each bar where the temperature of the whole assembly is raised through 500 c, Assume $E_s = 200\text{GPa}$, $E_a = 80\text{GPa}$, $\alpha_s = 12 \times 10^{-6} / ^\circ\text{C}$, $\alpha_a = 24 \times 10^{-6} / ^\circ\text{C}$	M1.03	A
OR			
2	A cylindrical steel pressure vessel 400 mm in diameter with a wall thickness of 20 mm, is subjected to an internal pressure of 4.5 MN/m ² . (a) Calculate the tangential and longitudinal stresses in the steel. (b) To what value may the internal pressure be increased if the stress in the steel is limited to 120 MN/m ² ?	M1.05	A
3	A cantilever of 4m long carries two point loads each 5 kN, one placed at free end and the other at 3 m from fixed end Draw SF and BM diagrams.	M2.02	A
OR			
4	Determine the support reactions 	M2.01	U
5	A stut 2.5 m long is 65 mm in diameter. One end of the shaft is fixed while its other end is hinged. Find the safe compressive load for the member using Euler's formula" allowing a factor of safety of 3.5. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$	M3.05	A
OR			
6	A solid shaft is to transmit 350 kW at 100 rpm. If the shear stress of shaft material is not to exceed 90 N/mm ² , find the diameter of the solid shaft.	M.3.02	A
7	Determine the support reactions	M 2.01	U

			
OR			
8	Explain Hooks law for ductile materials with neat diagram	M 1.02	U
9	Explain solid and hollow shaft based on strength	M3.01	U
OR			
10	Explain different types of beams with neat sketches	M2.01	U
11	Derive the expression for strain energy due to gradually applied load	M4.01	U
OR			
12	Explain the properties of wrought aluminium alloys	M 4.02	U